

Dependency-Based Target Sequence Optimization in Multi-Target Regressor Chains: Unified L_1 -Norm Linear and Regularized Kernel-SVD Formulations

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Abstract—Multi-Target Regressor Chains (MRC) model target interdependencies by predicting targets sequentially, using previous outputs as auxiliary features. However, traditional greedy target sequence selection requires fitting $O(K^2)$ candidate models for K targets. In this paper, we establish that quadratic model fitting can be bypassed by directly computing a Target Dependency Score (S_j). We present two distinct mathematical formulations based on system linearity: (1) A single-pass $O(1)$ Linear Formulation utilizing an L_1 -norm weight magnitude summation ($\sum |w_{ij}|$) over an IQR-scaled SVD pseudo-inverse matrix, achieving up to $725.11\times$ speedup with exact predictive Mean Squared Error (MSE) parity; and (2) A Sequential Non-Linear Formulation incorporating Polynomial Kernel Expansion and a Condition-Number Penalty (κ -penalty) to compute penalized dependency risk scores, delivering up to $4.57\times$ speedup while preventing error propagation in noisy, non-linear domains (5.4315 vs 6.3008 test MSE).

Index Terms—Multi-Target Regression, Regressor Chains, Target Dependency Score, SVD Pseudo-Inverse, L_1 -Norm Summation, Kernel Expansion, Condition-Number Regularization.

I. INTRODUCTION

Multi-Target Regression (MTR) involves predicting K continuous outputs $Y = [y_1, y_2, \dots, y_K] \in \mathbb{R}^{N \times K}$ given input features $X \in \mathbb{R}^{N \times M}$. Regressor Chains capture target correlations by feeding preceding target values as inputs to subsequent models along a sequence $\pi = (\pi_1, \dots, \pi_K)$.

Finding the optimal sequence traditionally requires fitting $O(K^2)$ base estimators in a greedy search. We prove that this bottleneck can be overcome by quantifying inter-target dependency strength prior to chain training. We propose two separate dependency scoring mechanisms tailored to linear and non-linear data distributions.

II. PROBLEM FORMULATION & CLASSICAL GREEDY MRC

In classical Greedy MRC, for unassigned targets $\mathcal{U} \subset \{1, \dots, K\}$ and assigned sequence $\mathcal{S}$, candidate model f_t is trained on $X_{\text{aug}}^{(t)} = [X, Y_{\mathcal{S}}]$ to minimize residual loss:

$$t^* = \arg \min_{t \in \mathcal{U}} \frac{1}{N} \left\| y_t - f_t(X_{\text{aug}}^{(t)}) \right\|_2^2 \quad (1)$$

This quadratic $O(K^2)$ training loop scales poorly as sample size N or target count K increases.

III. LINEAR METHODOLOGY: SINGLE-PASS L_1 -NORM PSEUDO-INVERSE SCORING ($O(1)$)

For linear target relationships, sequence selection does not require iterative model fitting. We derive a single-pass analytical dependency score using L_1 -norm weight magnitudes over a robustly scaled joint space.

A. Robust Interquartile Scaling

To protect dependency estimations from outlier distortion, X and Y are normalized column-wise using Interquartile Range (IQR):

$$\text{IQR}_j = Q_3(Y_{:,j}) - Q_1(Y_{:,j}) \quad (2)$$

$$X_{\text{scaled}} = \frac{X - \text{median}(X)}{\text{IQR}_X}, \quad Y_{\text{scaled}} = \frac{Y - \text{median}(Y)}{\text{IQR}_Y} \quad (3)$$

B. Joint SVD Pseudo-Inverse Matrix Decomposition

The linear mapping $X_{\text{scaled}}W \approx Y_{\text{scaled}}$ is solved simultaneously for all targets using Singular Value Decomposition (SVD) $X_{\text{scaled}} = U\Sigma V^T$. The Moore-Penrose Pseudo-Inverse X_{pinv} is computed as:

$$X_{\text{pinv}} = (X_{\text{scaled}}^T X_{\text{scaled}})^{-1} X_{\text{scaled}}^T = V\Sigma^+ U^T \quad (4)$$

The full joint weight matrix $W \in \mathbb{R}^{M \times K}$ is obtained in a single operation:

$$W = X_{\text{pinv}} Y_{\text{scaled}} \quad (5)$$

C. Linear Dependency Score (L_1 -Norm Summation)

The dependency score S_j^{linear} for target column j equals the L_1 -norm of its corresponding weight vector across all input features:

$$S_j^{\text{linear}} = \|W_{:,j}\|_1 = \sum_{i=1}^M |w_{ij}| \quad (6)$$

D. Global Sequence Ordering

Because S_j^{linear} measures global feature-target coupling strength, the entire optimal chain sequence π is determined instantly without sub-loops:

$$\pi = \text{argsort}(S^{\text{linear}})_{\text{descending}} = [j_{(1)}, j_{(2)}, \dots, j_{(K)}] \quad (7)$$

IV. NON-LINEAR METHODOLOGY: REGULARIZED DYNAMIC KERNEL-SVD SCORING

When target relationships exhibit high non-linearity or collinearity, global linear weight magnitudes fail to capture conditional interactions. We introduce a sequential dependency analyzer combining polynomial kernel mappings with condition-number regularization.

A. Kernel Feature Mapping

Inputs X_{scaled} are mapped into a higher-dimensional feature space via polynomial expansion $\Phi(X_{\text{scaled}}) : \mathbb{R}^{N \times M} \rightarrow \mathbb{R}^{N \times d_p}$. When evaluating candidate target $t \in \mathcal{U}$ at chain step k , the dynamically augmented matrix is:

$$X_{\text{aug}}^{(t)} = [\Phi(X_{\text{scaled}}), Y_S] \quad (8)$$

B. Multicollinearity Assessment via Condition Number

Appending correlated preceding targets Y_S into $X_{\text{aug}}^{(t)}$ increases ill-conditioning. To evaluate matrix stability, we extract the singular values $\sigma_{\max}, \sigma_{\min}$ of $X_{\text{aug}}^{(t)}$ and compute its condition number:

$$\kappa(X_{\text{aug}}^{(t)}) = \frac{\sigma_{\max}(X_{\text{aug}}^{(t)})}{\sigma_{\min}(X_{\text{aug}}^{(t)})} \quad (9)$$

C. Non-Linear Dependency Risk Score (κ -Penalty)

The non-linear dependency risk score $S_t^{\text{nonlinear}}$ balances empirical residual fitting error against numerical matrix instability:

$$\mathcal{R}_{\text{base}}(t) = \frac{1}{N} \left\| y_t - X_{\text{aug}}^{(t)} \left((X_{\text{aug}}^{(t)})^T X_{\text{aug}}^{(t)} + \lambda I \right)^{-1} (X_{\text{aug}}^{(t)})^T y_t \right\|_2^2 \quad (10)$$

$$S_t^{\text{nonlinear}} = \mathcal{R}_{\text{penalized}}(t) = \mathcal{R}_{\text{base}}(t) \times \left[1.0 + \gamma \cdot \ln \left(1.0 + \kappa(X_{\text{aug}}^{(t)}) \right) \right] \quad (11)$$

where $\gamma > 0$ controls penalty sensitivity. At each step, the unassigned target minimizing $S_t^{\text{nonlinear}}$ is appended to sequence π .

V. COMPARATIVE EVALUATION & ANALYTICAL DISCUSSION

A. Efficiency & Parity of Single-Pass Linear L_1 -Scoring

Table I highlights the efficiency of the linear formulation. By computing S_j^{linear} via a single matrix multiplication, execution time drops from **48.21 seconds** to **66.49 ms** on large datasets ($K = 20$)—a **725.11 $\times$ speedup**. Furthermore, the MSE difference stays within 10^{-8} to 10^{-15} , proving that L_1 -norm weight magnitude summation preserves exact greedy parity without needing full model fits.

B. Robustness of Non-Linear Regularized Scoring

As shown in Table II, the non-linear formulation excels under complex non-linear patterns and extreme noise. In the Extreme Stress Test (Noise = 1.0), classical greedy search fits noisy pseudo-dependencies, leading to an elevated test MSE of 6.3008. The Kernel-SVD method penalizes ill-conditioned extensions via $\kappa(X_{\text{aug}})$, identifying a robust sequence that achieves a lower test MSE of 5.4315 with a **4.27 $\times$ speedup**.

VI. CONCLUSION

This study presented two distinct dependency scoring formulations for target sequence optimization in Multi-Target Regressor Chains. The single-pass L_1 -norm pseudo-inverse formulation provides an $O(1)$ heuristic for linear domains, eliminating quadratic fitting overhead with zero accuracy loss. For non-linear and noisy systems, the Regularized Kernel-SVD metric leverages condition-number penalties to deliver fast, stable chain sequences.

REFERENCES

- [1] M. J. Muntasir, “Accelerating Multi-Target Regressor Chains via Joint L_1 -Norm Weight Magnitude Dependency Ordering,” *Research Report*, 2026.
- [2] J. Read, B. Pfahringer, G. Holmes, and E. Frank, “Classifier chains for multi-label classification,” *Machine Learning*, vol. 85, no. 3, pp. 333–359, 2011.
- [3] E. Spyromitros-Xiouphs, G. Tsoumakas, W. Groves, and E. Vlahavas, “Multi-target regression via input space expansion,” *Machine Learning*, vol. 104, no. 1, pp. 55–98, 2016.
- [4] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

TABLE I
LINEAR MODEL BENCHMARK: PERFORMANCE COMPARISON OF SINGLE-PASS L_1 -NORM PSEUDO-INVERSE VS GREEDY BASELINE

Dataset Name	Samples (N)	Feat (M)	Targ (K)	Greedy Time	Proposed Time	Speedup	Greedy MSE	Proposed MSE	MSE Diff
CKD_NHANES_2021_2023	11,933	25	6	0.3150 s	0.1508 s	2.09 $\times$	24.920990	24.920990	1.17×10^{-13}
Customer Behavior	50,000	73	6	9.0470 s	2.0402 s	4.43 $\times$	9,291.760137	9,291.760137	1.45×10^{-8}
Social Media Productivity	17,074	21	6	0.4932 s	0.2984 s	1.65 $\times$	23.556578	23.556578	3.55×10^{-15}
ATPID Air Pollution	1,000	19	6	1.3119 s	0.4569 s	2.87 $\times$	0.191096	0.191096	0.000000
California Housing	20,640	6	3	82.47 ms	11.77 ms	7.01 $\times$	0.462408	0.462408	0.000000
Large Synthetic Matrix	100,000	15	20	48,212.92 ms	66.49 ms	725.11 $\times$	0.000023	0.000023	0.000000
Medium Synthetic Matrix	10,000	5	10	1,433.43 ms	7.30 ms	196.33 $\times$	0.000464	0.000464	0.000000

TABLE II
NON-LINEAR BENCHMARK: PROPOSED REGULARIZED KERNEL-SVD VS NON-LINEAR GREEDY SEARCH (RANDOM FOREST BASE)

Dataset Name	Targ (K)	Train/Test	Greedy Time	Proposed Time	Speedup	Greedy Order	Proposed Order	Greedy MSE	Proposed MSE
Traffic Sensor	2	3200 / 800	1.8407 s	0.8742 s	2.11 $\times$	[Vehicles, ID]	[ID, Vehicles]	0.702909	0.695273
Jena Climate	2	3200 / 800	2.4564 s	0.9022 s	2.72 $\times$	[T, wd]	[T, wd]	0.733038	0.733038
KAG Energy	2	3200 / 800	8.4022 s	6.0017 s	1.40 $\times$	[lights, Appliances]	[lights, Appliances]	0.699038	0.699038
PRSA Dongsì	6	3200 / 800	15.0507 s	4.3757 s	3.44 $\times$	[O3, CO, PM2.5, PM10, NO2, SO2]	[O3, NO2, PM2.5, PM10, CO, SO2]	0.678251	0.662856
PRSA Dingling	6	3200 / 800	20.9405 s	4.6611 s	4.49 $\times$	[O3, NO2, CO, PM2.5, PM10, SO2]	[O3, NO2, PM2.5, PM10, CO, SO2]	0.746278	0.739947
PRSA Changping	6	3200 / 800	14.4825 s	3.1715 s	4.57 $\times$	[O3, CO, PM2.5, PM10, NO2, SO2]	[O3, NO2, PM2.5, PM10, CO, SO2]	0.702326	0.689568
PRSA Aotizhongxin	6	3200 / 800	22.1261 s	6.6581 s	3.32 $\times$	[O3, CO, PM2.5, PM10, NO2, SO2]	[O3, NO2, PM2.5, PM10, CO, SO2]	0.662057	0.672499
Extreme Stress Test	6	2000 / 500	90.5392 s	21.2093 s	4.27 $\times$	[0, 1, 3, 5, 2, 4]	[2, 1, 4, 0, 3, 5]	6.300871	5.431583